Non-Hermitian Lattice Sensing

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1 Non-Hermitian Quantum Sensing

1.1 Using Non-Hermitian Lattice Dynamics [1]

In this section, we will exploit the large uni-directional response generated by certain non-Hermitian lattice models for sensing purposes.

1.1.1 The Hatano-Nelson Model

The simplest relevant system to us is the **Hatano-Nelson Model**. This is a 1D tight-binding system of lattice sites, with asymmetric nearest-neighbour hoppings.
"Tight-binding" means the particles only hop to adjacent sites and not further.
"Asymmetric hoppings" mean that the wavefunction changes differently based on whether it hops to the left nearest neighbour or the right.
The Hamiltonian of the system is defined as follows

$$\hat{H} = iJ\sum_n (e^A|n+1\rangle\langle n| - e^{-A}|n\rangle\langle n+1|) \tag{1}$$

J defines the base amplitude scaling of the system, A is the amplification factor. $|n\rangle$ describes the lattice sides in the model (position eigenkets).
If $\hat{H}$ was Hermitian, the coefficients of the left-hopping and right-hopping terms would be conjugate symmetric. Thanks to the $e^{\pm A}$ factors, we can see that that is not the case here.
Now to obtain the corresponding Schrödinger equation for this Hamiltonian. We start with the Time-Dependent Schrödinger Equation (TDSE).

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$$

We'll set $\hbar = 1$ for simplicity. Now, we can express $|\psi(t)\rangle$ as a superposition of the position eigenkets

$$|\psi(t)\rangle = \sum_n \psi_n(t)|n\rangle$$

Where $\psi_n(t) = \langle n|\psi(t)\rangle$ is the complex probability amplitude of the particle being at position n. $|\psi_n(t)|^2$ would be the probability of it. Now multiplying the TDSE with $\langle n|$ from the left, we get

$$i\frac{d}{dt}\langle n|\psi(t)\rangle = \langle n|\hat{H}|\psi(t)\rangle$$

$$i\dot{\psi}_n = \langle n|\hat{H}\sum_m \psi_m|m\rangle$$

$$\cancel{i}\dot{\psi}_n = \langle n|\Big[\cancel{i}J\sum_k (e^A|k+1\rangle\langle k| - e^{-A}|k\rangle\langle k+1|\Big]\sum_m \psi_m|m\rangle$$

Using the orthonormality of the eigenkets ($\langle n|m\rangle = \delta_{n,m}$), we get

$$\dot{\psi}_n = J\sum_{k,m}\psi_m(e^A\delta_{n,k+1}\delta_{k,m} - e^{-A}\delta_{n,k}\delta_{k+1,m})$$

$$\dot{\psi}_n = J\sum_k \psi_k e^A\delta_{n,k+1} - \psi_{k+1}e^{-A}\delta_{n,k}$$

$$\therefore \dot{\psi}_n = Je^{A}\psi_{n-1} - Je^{-A}\psi_{n+1} \tag{2}$$

In steady state, $\dot{\psi}_n = 0$, so the equation becomes

$$\cancel{J}e^{A}\psi_{n-1} = \cancel{J}e^{-A}\psi_{n+1}$$

$$\Rightarrow \frac{\psi_{n+1}}{\psi_{n-1}} = e^{2A}$$

Assuming a uniform growth rate across sites, we get

$$\psi_{n+1} = e^{A}\psi_n \tag{3}$$

Hence, this is a system where a wavefunction's amplitude grows by e^A every time a particle hops to the right, and decays by e^{-A} every time it hops to the left. This happens regardless of the initial energy.

With this picture in mind, let us talk about the *susceptibility*, given by the single-particle Green's function $\chi(n, m; t)$. It answers the question: If a particle is placed at site m at $t = 0$, what is the resultant amplitude of the wavefunction at site n, at time t? In other words, what is the probability amplitude that it will be at site n at time t? Using the time evolution relation $|m(t)\rangle = e^{-i\hat{H}t}|m\rangle$, this function is given by

$$\chi(n, m; t) = \langle n|m(t)\rangle = \langle n|e^{-i\hat{H}t}|m\rangle \tag{4}$$

But from our understanding of the system, it is easy to see that

$$\chi(n, m; t) = e^{A(n-m)}\chi_0(n, m; t) \tag{5}$$

where $\chi_0(n, m; t)$ represents the Green function for a Hatano-Nelson chain where $A = 0$, i.e., a *Hermitian* tight-binding chain. We can understand this by saying that the wavefunction amplitude for the non-Hermitian chain will be $e^{A(n-m)}$ times larger than if it were a Hermitian chain.

1.1.2 Construction of the Sensor Model

Now it is clear to us that the Hatano-Nelson chain breaks reciprocity for any $A \neq 0$. We can restore this symmetry by having a system with two uncoupled Hatano-Nelson chains, indexed by $\sigma = \{\uparrow, \downarrow\}$ with amplification factors $A = \{A_\uparrow, A_\downarrow\}$. If we choose the factors such that $A_\uparrow = -A_\downarrow$, then the system as a whole recovers the lost symmetry. We say that these 2 chains have opposite "chirality". We wish to break the symmetry of this system with a parameter whose magnitude we want to estimate. But for now, let us focus on how to realise this model.

We consider an N-site chain of driven, coupled bosonic modes, described by the Hermitian Hamiltonian

$$\hat{H}_B = \sum_{n=1}^{N} \left(iw\hat{a}^\dagger_{n+1}\hat{a}_n + i\Delta\hat{a}^\dagger_{n+1}\hat{a}^\dagger_n + h.c. \right) \tag{6}$$

- "Driven" means the system is energised by an external power source.
- "Coupled" means the lattice sites are related in some way.
- The lattice sites are occupied by bosons (specifically photons), and so their behaviour at each lattice site j is described by annihilation operators $\hat{a}^\dagger_j$.
- w is the nearest-neighbour hopping term. It is a measure of the likelihood that a photon will disappear from site $n + 1$ and appear in site n.
- Δ is the nearest-neighbour two-photon drive. It is a measure of the likelihood that a pair of photons across neighbouring sites is created or destroyed. Recall that photons are massless and thus can be created and destroyed. This obviously is an active process, requiring external power (for creation of photons). That is why the system must be "driven".
- *h.c.* stands for Hermitian Conjugate. It is written in place of the hermitian conjugates of the terms written before it, which in this case would be $-iw\hat{a}^\dagger_n\hat{a}_{n+1}$ and $-i\Delta\hat{a}_n\hat{a}_{n+1}$.
- We take both w and Δ to be real and positive, and $w > \Delta$.

This model describes a 1D cavity array subject to parametric drives on each bond.
Now let us move from the basis of annihilation operators to the basis of *Hermitian quadrature operators* $\hat{x}_j$ and $\hat{p}_j$ (local and canonical). They are defined via

$$\hat{a}_j = \frac{1}{\sqrt{2}}(\hat{x}_j + i\hat{p}_j) \qquad \hat{a}_j^\dagger = \frac{1}{\sqrt{2}}(\hat{x}_j - i\hat{p}_j) \tag{7}$$

$$\Rightarrow \hat{x}_j = \frac{1}{\sqrt{2}}(\hat{a}_j + \hat{a}_j^\dagger) \qquad \hat{p}_j = \frac{1}{i\sqrt{2}}(\hat{a}_j - \hat{a}_j^\dagger) \tag{8}$$

Thus our Hamiltonian becomes

$$\hat{H}_B = \frac{1}{2}\sum_{n=1}^{N}[iw(\hat{x}_{n+1}-i\hat{p}_{n+1})(\hat{x}_n+i\hat{p}_n)+i\Delta(\hat{x}_{n+1}-i\hat{p}_{n+1})(\hat{x}_n-i\hat{p}_n)-iw(\hat{x}_n-i\hat{p}_n)(\hat{x}_{n+1}+i\hat{p}_{n+1})-i\Delta(\hat{x}_n+i\hat{p}_n)(\hat{x}_{n+1}+i\hat{p}_{n+1})]$$

$$\hat{H}_B = \frac{1}{2}\sum_{n=1}^{N}\Big[iw([\hat{x}_{n+1},\hat{x}_n]+[\hat{p}_{n+1},\hat{p}_n]+i\{\hat{x}_{n+1},\hat{p}_n\}-i\{\hat{p}_{n+1},\hat{x}_n\})+i\Delta([\hat{x}_{n+1},\hat{x}_n]-[\hat{p}_{n+1},\hat{p}_n]-i\{\hat{x}_{n+1},\hat{p}_n\}+i\{\hat{p}_{n+1},\hat{x}_n\})\Big]$$

From known commutation relations of annihilation operators ($[\hat{a}_i,\hat{a}_j]=[\hat{a}_i^\dagger,\hat{a}_j^\dagger]=0,\ [\hat{a}_i,\hat{a}_j^\dagger]=\delta_{ij}$), we can find that

$$[\hat{x}_i,\hat{x}_j]=0 \quad [\hat{p}_i,\hat{p}_j]=0 \quad [\hat{x}_i,\hat{p}_j]=i\delta_{ij}$$

Thus, the Hamiltonian reduces to

$$\hat{H}_B = \frac{1}{2}\sum_{n=1}^{N}[-w(2\hat{x}_{n+1}\hat{p}_n - 2\hat{p}_{n+1}\hat{x}_n) - \Delta(-2\hat{x}_{n+1}\hat{p}_n + 2\hat{p}_{n+1}\hat{x}_n)]$$

$$\hat{H}_B = \sum_{n=1}^{N}\big(-(w-\Delta)\hat{x}_{n+1}\hat{p}_n + (w+\Delta)\hat{p}_{n+1}\hat{x}_n\big) \tag{9}$$

One can see the similarities between this equation and Eq.(1). Using similar methods as before, we can obtain the equations of motion:

$$\dot{\hat{x}}_n = Je^{A}\hat{x}_{n-1} - Je^{-A}\hat{x}_{n+1} \tag{10}$$

$$\dot{\hat{p}}_n = Je^{-A}\hat{p}_{n-1} - Je^{A}\hat{p}_{n+1} \tag{11}$$

Where the hopping amplitude J and amplification factor A are defined as follows

$$J \equiv \sqrt{w^2-\Delta^2} \qquad , \qquad e^{2A} \equiv \frac{w+\Delta}{w-\Delta} \Rightarrow A \equiv \frac{1}{2}\ln\left(\frac{w+\Delta}{w-\Delta}\right)$$

(You can now see why we enforced that $w > \Delta$, so that J and A will be defined.)
From this, we can see the importance of the external driving term Δ for causing the system's non-Hermiticity. If $\Delta = 0$, we can see that J is simply equal to w, and that amplification factor $A = 0$. So the Hamiltonian becomes Hermitian.

Thus, we can see that this Bosonic chain is mathematically equivalent to the double Hatano-Nelson chain we desire. The complex wavefunctions from before have been replaced by Hermitian quadratures, but this will make no difference.

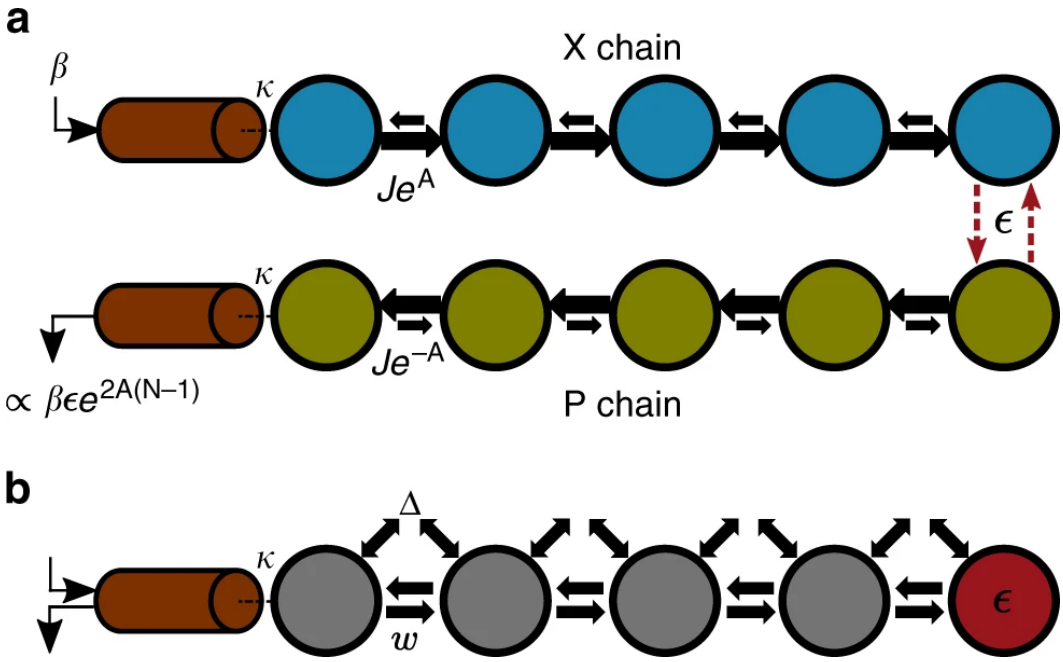


Figure 1: (a) The two Hatano-Nelson chains, coupled by parameter ϵ, (b) A realisation of (a), using a coupled driven chain of bosonic modes

1.1.3 Introducing the Waveguide [2]

To probe our lattice, we couple it to an input-output waveguide, as seen in Fig.1. In this section, we will explain the extra terms this waveguide adds to the Hamiltonian, seen in Eq.(9) of [1].

NOTE: This section has long winded derivations for these extra terms. You can skip to (47) *and* (53) *to see the conclusions of these derivations*

With the introduction of the waveguide, we can regard this setup as an *Open Quantum System* with Hamiltonian given by

$$\hat{H} = \hat{H}_{\text{sys}} + \hat{H}_{\text{bath}} + \hat{H}_{\text{int}} \tag{12}$$

In our case, it would be

$$\hat{H} = \hat{H}_{\text{cavity}} + \hat{H}_{\text{waveguide}} + \hat{H}_{\text{coupling}} \tag{13}$$

- $\hat{H}_{\text{cavity}} \equiv \hat{H}_{\text{sys}}$ will be equal to $\hat{H}_B$ + (perturbation), which we will deal with in the next part.
- Since a waveguide has an infinite continuum of modes (continuous energy spectrum), its Hamiltonian $\hat{H}_{\text{waveguide}} \equiv \hat{H}_{\text{bath}}$ will be a superposition of harmonic oscillators at every frequency.

$$\hat{H}_{\text{bath}} = \hbar \int_{-\infty}^{\infty} d\omega\, \omega\, \hat{b}_\omega^\dagger\, \hat{b}_\omega \tag{14}$$

  where $\hat{b}_\omega$ are the boson annihilation operators for the bath. They are continuous, unlike the system annihilation operators. They follow the commutation relation

$$[\hat{b}_\omega, \hat{b}_{\omega'}^\dagger] = \delta(\omega - \omega') \tag{15}$$

- The interaction/coupling Hamiltonian $\hat{H}_{\text{int}} \equiv \hat{H}_{\text{coupling}}$ comes from the bilinear coupling of the 1st cavity to the continuum of waveguide modes.

$$\hat{H}_{\text{int}} = \hbar \int_{-\infty}^{\infty} d\omega\, g(\omega)(\hat{b}_\omega + \hat{b}_\omega^\dagger)(\hat{a}_1 + \hat{a}_1^\dagger) \tag{16}$$

  We use the *Rotating Wave Approximation* (RWA) to drop the terms $\hat{b}_\omega \hat{a}_1$ and $\hat{b}_\omega^\dagger \hat{a}_1^\dagger$. So we get

$$\hat{H}_{\text{int}} = \hbar \int_{-\infty}^{\infty} d\omega\, g(\omega)[\hat{b}_\omega \hat{a}_1^\dagger + \hat{b}_\omega^\dagger \hat{a}_1] \tag{17}$$

Rotating Wave Approximation:
The annihilation operators have their own time evolution. The cavity oscillates at ω_c and the waveguide oscillates at ω.

$$\hat{a}_1 \to \hat{a}_1 e^{-i\omega_c t} \qquad \hat{b}_\omega \to \hat{b}_\omega e^{-i\omega t}$$

Now it is simple for us to see that the "like" terms (i.e. both with or both without dagger) will operate at frequency $\pm(\omega_c + \omega)$ and the "cross" terms will operate at frequency $\pm(\omega_c - \omega)$. Since the timescale will be set by the slower (cross) terms, the faster (like) terms will be oscillating so fast that they average to zero. Thus, in our timescale, they can be ignored, leading to this approximation.

We will also use the *Markov Approximation* on our $g(\omega)$ term, saying that its dependence on frequency is negligible and it can be written as

$$g(\omega) \approx \sqrt{\frac{\kappa}{2\pi}} \tag{18}$$

This approximation is due to similar reasons as the RWA, in that the timescale of the waveguide is much smaller than the timescale of the system, or $\omega \gg |\omega - \omega_c|$. Hence, differences in ω don't really matter in the scheme of things, so the coupling factor can be approximated to be constant.
This specific form is chosen for mathematical convenience that we will see later.
Thus, the interaction Hamiltonian is

$$\hat{H}_{\text{int}} = \hbar\sqrt{\frac{\kappa}{2\pi}} \int_{-\infty}^{\infty} d\omega \, [\hat{b}_\omega \hat{a}_1^\dagger + \hat{b}_\omega^\dagger \hat{a}_1] \tag{19}$$

Now that we have our complete Hamiltonian, we can use the Heisenberg equation of motion shown below

$$\dot{\hat{A}} = \frac{d\hat{A}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{A}] + \frac{\partial \hat{A}}{\partial t} \tag{20}$$

Using this, we can derive equations of motion for $\hat{b}_\omega$ and $\hat{a}_1$. The partial derivative term will be 0 for both operators. They have no explicit time dependence.

For $\hat{b}_\omega$:
We get

$$\dot{\hat{b}}_\omega = \frac{i}{\hbar}[\hat{H}, \hat{b}_\omega] \tag{21}$$

We'll need to find the commutator of $\hat{b}_\omega$ with all 3 Hamiltonians mentioned in Eq.(12). Since all of them have $\hbar$ that will cancel with the denominator, we can ignore the $\hbar$ in our equations below.
For $\hat{H}_{\text{sys}}$, refer Eq.(6). Ignore the perturbation for this part. Now, we can use the fact that cavity and waveguide operators commute (separate sytems), to see that

$$[\hat{H}_{\text{sys}}, \hat{b}_\omega] = 0 \tag{22}$$

For $\hat{H}_{\text{bath}}$,

$$\begin{aligned}
[\hat{H}_{\text{bath}}, \hat{b}_\omega] &= \left[\int d\omega' \, \omega' \, \hat{b}_{\omega'}^\dagger \, \hat{b}_{\omega'}, \hat{b}_\omega \right] \\
&= \int d\omega' \, \omega' \, [\hat{b}_{\omega'}^\dagger \, \hat{b}_{\omega'}, \hat{b}_\omega] \\
&= \int d\omega' \, \omega' \Big(\hat{b}_{\omega'}^\dagger [\hat{b}_{\omega'}, \hat{b}_\omega] + [\hat{b}_{\omega'}^\dagger, \hat{b}_\omega] \hat{b}_{\omega'} \Big) \\
&= \int d\omega' \, \omega' \Big(0 - \delta(\omega - \omega') \hat{b}_{\omega'} \Big) \\
[\hat{H}_{\text{bath}}, \hat{b}_\omega] &= -\omega \, \hat{b}_\omega
\end{aligned} \tag{23}$$

For $\hat{H}_{\text{int}}$,

$$\begin{aligned}
[\hat{H}_{\text{int}}, \hat{b}_\omega] &= \left[\sqrt{\frac{\kappa}{2\pi}} \int d\omega' \, (\hat{b}_{\omega'} \hat{a}_1^\dagger + \hat{b}_{\omega'}^\dagger \hat{a}_1), \hat{b}_\omega\right] \\
&= \sqrt{\frac{\kappa}{2\pi}} \int d\omega' \Big([\hat{b}_{\omega'} \hat{a}_1^\dagger, \hat{b}_\omega] + [\hat{b}_{\omega'}^\dagger \hat{a}_1, \hat{b}_\omega] \Big) \\
&= \sqrt{\frac{\kappa}{2\pi}} \int d\omega' \Big([\hat{b}_{\omega'}, \hat{b}_\omega] \hat{a}_1^\dagger + \hat{b}_{\omega'} [\hat{a}_1^\dagger, \hat{b}_\omega] + [\hat{b}_{\omega'}^\dagger, \hat{b}_\omega] \hat{a}_1 + \hat{b}_{\omega'}^\dagger [\hat{a}_1, \hat{b}_\omega] \Big) \\
&= \sqrt{\frac{\kappa}{2\pi}} \int d\omega' \Big(0 + 0 - \delta(\omega - \omega') \hat{a}_1 + 0 \Big) \\
[\hat{H}_{\text{int}}, \hat{b}_\omega] &= -\sqrt{\frac{\kappa}{2\pi}} \hat{a}_1
\end{aligned} \tag{24}$$

Therefore, the equation of motion for $\hat{b}_\omega$ is

$$\begin{aligned}
\dot{\hat{b}}_\omega &= i[\hat{H}, \hat{b}_\omega] \\
\dot{\hat{b}}_\omega &= -i\omega \, \hat{b}_\omega - i\sqrt{\frac{\kappa}{2\pi}} \hat{a}_1
\end{aligned} \tag{25}$$

For $\hat{a}_1$:
Ignoring $\hbar$ like before, we check its commutator for each Hamiltonian.
For $\hat{H}_{\text{sys}}$, We should only consider the first cavity part, since for all other cavities, the commutator will be zero.

$$\begin{aligned}
[\hat{H}_{\text{sys}}, \hat{a}_1] &= [\omega_c \hat{a}_1^\dagger \hat{a}_1, \hat{a}_1] \\
&= \omega_c \big(\hat{a}_1^\dagger [\hat{a}_1, \hat{a}_1] + [\hat{a}_1^\dagger, \hat{a}_1] \hat{a}_1 \big) \\
&= \omega_c \big(0 - \hat{a}_1 \big) \\
[\hat{H}_{\text{sys}}, \hat{a}_1] &= -\omega_c \hat{a}_1
\end{aligned} \tag{26}$$

For $\hat{H}_{\text{bath}}$, it will only contain commutators between waveguide operators and cavity operators, so we can safely say

$$[\hat{H}_{\text{bath}}, \hat{a}_1] = 0 \tag{27}$$

For $\hat{H}_{\text{int}}$,

$$\begin{aligned}
[\hat{H}_{\text{int}}, \hat{a}_1] &= \left[\sqrt{\frac{\kappa}{2\pi}} \int d\omega' \, (\hat{b}_{\omega'} \hat{a}_1^\dagger + \hat{b}_{\omega'}^\dagger \hat{a}_1), \hat{a}_1\right] \\
&= \sqrt{\frac{\kappa}{2\pi}} \int d\omega' \Big([\hat{b}_{\omega'} \hat{a}_1^\dagger, \hat{a}_1] + [\hat{b}_{\omega'}^\dagger \hat{a}_1, \hat{a}_1] \Big) \\
&= \sqrt{\frac{\kappa}{2\pi}} \int d\omega' \Big([\hat{b}_{\omega'}, \hat{a}_1] \hat{a}_1^\dagger + \hat{b}_{\omega'} [\hat{a}_1^\dagger, \hat{a}_1] + [\hat{b}_{\omega'}^\dagger, \hat{a}_1] \hat{a}_1 + \hat{b}_{\omega'}^\dagger [\hat{a}_1, \hat{a}_1] \Big) \\
&= \sqrt{\frac{\kappa}{2\pi}} \int d\omega' \Big(0 - \hat{b}_{\omega'} + 0 + 0 \Big) \\
[\hat{H}_{\text{int}}, \hat{a}_1] &= -\sqrt{\frac{\kappa}{2\pi}} \int d\omega' \hat{b}_{\omega'} = -\sqrt{\frac{\kappa}{2\pi}} \int d\omega \, \hat{b}_\omega
\end{aligned} \tag{28}$$

Therefore, the equation of motion for $\hat{a}_1$ is

$$\begin{aligned}
\dot{\hat{a}}_1 &= i[\hat{H}, \hat{a}_1] \\
\dot{\hat{a}}_1 &= -i\omega_c \hat{a}_1 - i\sqrt{\frac{\kappa}{2\pi}} \int d\omega \, \hat{b}_\omega
\end{aligned} \tag{29}$$

Now we have our equations of motion (EOMs) (25) and (29) for the waveguide and cavity operators respectively, we need to solve them. Start with the waveguide EOM (25).

$$\begin{aligned}
\dot{\hat{b}}_\omega(t) &= -i\omega \, \hat{b}_\omega(t) - i\sqrt{\frac{\kappa}{2\pi}} \hat{a}_1(t) \\
\Rightarrow \dot{\hat{b}}_\omega(t) + i\omega \, \hat{b}_\omega(t) &= -i\sqrt{\frac{\kappa}{2\pi}} \hat{a}_1(t)
\end{aligned} \tag{30}$$

This is standard form of a Linear Differential Equation. We can solve this by finding the integrating factor, in this case $e^{i\omega t}$, multiply both sides by it and integrate from 0 to t. Doing this, we obtain the solution

$$\hat{b}_\omega(t) = e^{-i\omega t}\,\hat{b}_\omega(0) - i\sqrt{\frac{\kappa}{2\pi}}\int_0^t e^{-i\omega(t-t')}\hat{a}_1(t')\,dt' \tag{31}$$

We can substitute this in (29) to get

$$\begin{aligned}
\dot{\hat{a}}_1(t) &= -i\omega_c\hat{a}_1(t) - i\sqrt{\frac{\kappa}{2\pi}}\int d\omega\, e^{-i\omega t}\,\hat{b}_\omega(0) - \frac{\kappa}{2\pi}\int d\omega \int_0^t dt'\, e^{-i\omega(t-t')}\hat{a}_1(t') \\
&= -i\omega_c\hat{a}_1(t) - i\sqrt{\frac{\kappa}{2\pi}}\int d\omega\, e^{-i\omega t}\,\hat{b}_\omega(0) - \frac{\kappa}{\cancel{2\pi}}\int_0^t dt'\,\cancel{2\pi}\delta(t-t')\,\hat{a}_1(t') \\
\dot{\hat{a}}_1(t) &= -i\omega_c\hat{a}_1(t) - \frac{\kappa}{2}\hat{a}_1(t) - i\sqrt{\frac{\kappa}{2\pi}}\int d\omega\, e^{-i\omega t}\,\hat{b}_\omega(0)
\end{aligned} \tag{32}$$

The 1/2 factor for the second term of the RHS comes from the fact that $\delta(t-t')$ reaches its peak at $t'=t$, which is at the boundary of the integration. Therefore, only half of the full integration is included in the integral.
The last term on the RHS can be thought of as the contribution of the "input field". This is because it only contains waveguide terms and integrates its contributions over all frequencies that the waveguide can cover. Thus, we define the input field term

$$\hat{b}_{\text{in}}(t) = \frac{1}{\sqrt{2\pi}}\int d\omega\, e^{-i\omega t}\,\hat{b}_\omega(0) \tag{33}$$

The $1/\sqrt{2\pi}$ factor is from Fourier convention. This term follows the commutation relation given below.

$$[\hat{b}_{\text{in}}(t), \hat{b}^\dagger_{\text{in}}(t')] = \delta(t-t') \tag{34}$$

Putting this new term into (32), we get

$$\dot{\hat{a}}_1(t) = -i\omega_c\hat{a}_1(t) - \frac{\kappa}{2}\hat{a}_1(t) - i\sqrt{\kappa}\,\hat{b}_{\text{in}}(t) \tag{35}$$

Now to solve this.
Till now we've said that the waveguide can provide any frequency. Now we'll fix the input "driving" frequency at ω_d. To clarify, the waveguide still has an infinite continuum of modes, but only the ω_d mode will be carrying photons for us. This means operator $\hat{a}_1(t)$ oscillates at frequency $\omega_c + \omega_d$.
We will now move into a rotating frame of reference, which is rotating at frequency ω_d. The corresponding cavity operator will be similar to $\hat{a}_1(t)$, just with the ω_d factored out

$$\tilde{a}_1(t) = \hat{a}_1(t)e^{i\omega_d t} \tag{36}$$

$$\Rightarrow \hat{a}_1(t) = \tilde{a}_1(t)e^{-i\omega_d t} \tag{37}$$

Substituting this into (35), we get

$$\begin{aligned}
\dot{\tilde{a}}_1(t)e^{-i\omega_d t} - i\omega_d\,\tilde{a}_1(t)e^{-i\omega_d t} &= -i\omega_c\tilde{a}_1(t)e^{-i\omega_d t} - \frac{\kappa}{2}\tilde{a}_1(t)e^{-i\omega_d t} - i\sqrt{\kappa}\,\hat{b}_{\text{in}}(t) \\
\Rightarrow \dot{\tilde{a}}_1(t)e^{-i\omega_d t} &= -i(\omega_c-\omega_d)\tilde{a}_1(t)e^{-i\omega_d t} - \frac{\kappa}{2}\tilde{a}_1(t)e^{-i\omega_d t} - i\sqrt{\kappa}\,\hat{b}_{\text{in}}(t)
\end{aligned} \tag{38}$$

Now in our case, we take this driving frequency to be equal to the cavity's resonance frequency. That is, $\omega_d = \omega_c$. Thus in our rotating frame, the resonant frequency of the cavity is zero.
Our equation now becomes

$$\begin{aligned}
\dot{\tilde{a}}_1(t)e^{-i\omega_c t} &= -\frac{\kappa}{2}\tilde{a}_1(t)e^{-i\omega_c t} - i\sqrt{\kappa}\,\hat{b}_{\text{in}}(t) \\
\dot{\tilde{a}}_1(t) &= -\frac{\kappa}{2}\tilde{a}_1(t) - i\sqrt{\kappa}\,\hat{b}_{\text{in}}(t)e^{i\omega_c t}
\end{aligned} \tag{39}$$

Now, since the waveguide is carrying a classical tone at ω_d, we can write it as

$$\hat{b}_{\text{in}}(t) = \beta e^{-i\omega_d t} = \beta e^{-i\omega_c t} \tag{40}$$

where $\beta = |\beta|e^{i\theta}$ is the complex amplitude of the drive. Substituting this into (39),

$$\dot{\tilde{a}}_1(t) = -\frac{\kappa}{2}\tilde{a}_1(t) - i\sqrt{\kappa}\,\beta \tag{41}$$

Now, referring to the Heisenberg equation of motion (20), we know that the terms in the RHS of (41) should come from $\frac{i}{\hbar}[\hat{H}, \tilde{a}_1(t)]$. So we can guess and verify the Hamiltonian from this.
By moving into the rotating frame, the $-i\omega_c \hat{a}_1(t)$ term got cancelled, which was the contribution of $\hat{H}_{\text{cavity}}$. Therefore, the terms in the RHS of (41) are only from $\hat{H}_{\text{waveguide}}$ and $\hat{H}_{\text{coupling}}$.
The first RHS term is the damping term, with a damping rate of κ. We can say this comes from an unspecified operator $\hat{H}_\kappa$, which represents the damping at the first site due to coupling. We've separated out the damping part, now we need to find the driving part. As we did before, we ignore $\hbar$.

$$\not{\hbar}[\hat{H}_{\text{drive}}, \tilde{a}_1] = -\not{\hbar}\sqrt{\kappa}\,\beta \tag{42}$$

We can only guess possible Hamiltonians and verify them by plugging them into the equation. Let us try

$$\hat{H}_{\text{drive}} = \alpha \tilde{a}_1^\dagger + \alpha^* \tilde{a}_1 \tag{43}$$

Plugging this in, we get

$$\begin{aligned} [\hat{H}_{\text{drive}}, \tilde{a}_1] &= -\sqrt{\kappa}\,\beta \\ \alpha[\tilde{a}_1^\dagger, \tilde{a}_1] + \alpha^*[\tilde{a}_1, \tilde{a}_1] &= -\sqrt{\kappa}\,\beta \\ -\alpha &= -\sqrt{\kappa}\,\beta \\ \alpha &= \sqrt{\kappa}\,\beta \end{aligned} \tag{44}$$

Therefore,

$$\hat{H}_{\text{drive}} = \sqrt{\kappa}\,\beta\,\tilde{a}_1^\dagger + \sqrt{\kappa}\,\beta^*\,\tilde{a}_1 \tag{45}$$

Now, to match with the equation in [1], we need to modify the phase of β and make the substitution $\beta \to -i\beta$. This is possible due to the fact that β is a complex number with arbitrary phase θ, so we can change it as we like.
Thus, the new Hamiltonian is

$$\begin{aligned} \hat{H}_{\text{drive}} &= -i\sqrt{\kappa}\,\beta\,\tilde{a}_1^\dagger + i\sqrt{\kappa}\,\beta^*\tilde{a}_1 \\ &= -i\sqrt{\kappa}(\beta\,\tilde{a}_1^\dagger - \beta^*\tilde{a}_1) \\ \hat{H}_{\text{drive}} &= -i\sqrt{\kappa}(\beta\,\tilde{a}_1^\dagger - \text{h.c.}) \end{aligned} \tag{46}$$

Now, the full Hamiltonian (unperturbed) can be written out as follows.

$$\boxed{\hat{H} = \hat{H}_B + \hat{H}_\kappa - i\sqrt{\kappa}(\beta\,\tilde{a}_1^\dagger - \text{h.c.})} \tag{47}$$

This includes the effects of coupling the waveguide to the lattice chain. To recap, $\hat{H}_B$ is the Hamiltonian of the Bosonic chain, $\hat{H}_\kappa$ describes damping at site 1 due to coupling of the waveguide, κ is the corresponding damping rate, β is the complex amplitude of the drive provided by the waveguide, and $\hat{a}_1$ is the annihilation operator of the first bosonic cavity.

Our next aim is to get the output field operator $\hat{b}_{\text{out}}(t)$. We start in a similar way to how we derived the input operator. Solve Eq.(30), but this time, integrate backward from t_1 to t ($t_1 > t$) instead of forward from 0 to t.

$$\hat{b}_\omega(t) = e^{-i\omega(t-t_1)}\,\hat{b}_\omega(0) + i\sqrt{\frac{\kappa}{2\pi}}\int_t^{t_1} e^{-i\omega(t-t')}\hat{a}_1(t')\,dt' \tag{48}$$

Let's understand why we do this. Let t be the instant the waveguide's field interacts with the cavity. The evolution of the field before that instant would be the input, hence why $\hat{b}_{\text{in}}(t)$ was calculated by integrating from 0 to t. The evolution of the field after that instant would in turn be the output, hence why we are doing this integration. Now we define the output field operator, similar to Eq.(33), as

$$\hat{b}_{\text{out}}(t) = \frac{1}{\sqrt{2\pi}}\int d\omega\, e^{-i\omega(t-t_1)}\,\hat{b}_\omega(t_1) \tag{49}$$

Now substituting (48) into (29) and evaluating like earlier, we get

$$\begin{aligned} \dot{\hat{a}}_1(t) &= -i\omega_c\hat{a}_1(t) + \frac{\kappa}{2}\hat{a}_1(t) - i\sqrt{\frac{\kappa}{2\pi}}\int d\omega\, e^{-i\omega(t-t_1)}\,\hat{b}_\omega(t_1) \\ \Rightarrow \dot{\hat{a}}_1(t) &= -i\omega_c\hat{a}_1(t) + \frac{\kappa}{2}\hat{a}_1(t) - i\sqrt{\kappa}\,\hat{b}_{\text{out}}(t) \end{aligned} \tag{50}$$

This is the *time-reversed* equation of motion.
Comparing this to (35), we see they are related by the following substitutions:

$$\hat{b}_{\text{in}}(t) \to \hat{b}_{\text{out}}(t)$$
$$\sqrt{\kappa} \to \sqrt{\kappa}$$
$$-\frac{\kappa}{2}\hat{a}_1(t) \to \frac{\kappa}{2}\hat{a}_1(t)$$

Subtracting (35) from (50), we get

$$0 = \kappa\,\hat{a}_1(t) - i\sqrt{\kappa}[\hat{b}_{\text{out}}(t) - \hat{b}_{\text{in}}(t)]$$
$$\Rightarrow \hat{b}_{\text{out}}(t) - \hat{b}_{\text{in}}(t) = -i\sqrt{\kappa}\,\hat{a}_1(t) \tag{51}$$

The $-i$ factor in the RHS is ultimately an overall phase, which can be ignored, as it does not change our observables. So the input-output relation becomes

$$\hat{b}_{\text{out}}(t) - \hat{b}_{\text{in}}(t) = \sqrt{\kappa}\,\hat{a}_1(t) \tag{52}$$

We can break the input into two parts: the input tone β and the noise, which we have neglected till now. To match the notation in [1], we can rewrite the input-output relation to finally get

$$\boxed{\hat{B}^{(\text{out})}(t) = \Big(\beta + \hat{B}^{(\text{in})}(t)\Big) + \sqrt{\kappa}\,\hat{a}_1(t)} \tag{53}$$

1.1.4 Perturbing the System

Now that the setup is done, we need to use it to estimate a parameter. We apply a Hermitian perturbation $\epsilon\hat{V}$, where $\hat{V}$ is some system operator. Our goal is to estimate ϵ.
Therefore, our final full Hamiltonian, including the perturbation and coupling effects from the waveguide, is

$$\boxed{\hat{H}[\epsilon] = \hat{H}_B + \epsilon\hat{V} + \hat{H}_\kappa - i\sqrt{\kappa}(\beta\,\tilde{a}_1^\dagger - \text{h.c.})} \tag{54}$$

1.1.5 Measuring

Our goal is to estimate ϵ by making an optimal measurement of the output field $\hat{B}^{(\text{out})}$. We'll take ϵ to have units of frequency, and $\hat{V}$ to be dimensionless.
Moreover, we'll make the safe assumption that ϵ is very small – small enough that we can only estimate it by integrating the output field over a long time interval τ.
Say we turn on the drive tone at $t = 0$. Then, the integral of the output field from 0 to τ can be written as an operator, known as a *temporal mode*.

$$\hat{\mathcal{B}}_\tau(N) = \frac{1}{\sqrt{\tau}}\int_0^\tau dt\,\hat{B}^{(\text{out})}(t) \tag{55}$$

The prefactor $1/\sqrt{\tau}$ is chosen for normalisation purposes, so that the operator satisfies

$$[\hat{\mathcal{B}}_\tau(N), \hat{\mathcal{B}}_\tau^\dagger(N)] = 1 \tag{56}$$

This allows it to be regarded as a bosonic lowering operator. We will be measuring this operator to try and get the information we need to estimate ϵ. We write an explicit dependence on N as that dependence will be of great interest to us.
The maximum amount of information available in $\hat{\mathcal{B}}_\tau(N)$ is given by its *Quantum Fisher Information* (QFI). However, as we've learnt before, QFI quantifies the information available, assuming an optimal measurement has been performed. It is up to us to find out what that optimal measurement is. Let's do that.
We start by declaring that the system we're working with is *linear* and *Gaussian*. It is linear due to the form of its equations of motion (10) and (11). They are linear, so the system is linear. The system is Gaussian because it is driven by a coherent tone of amplitude β. Coherent tones, such as light from a laser, form coherent states, which are Gaussian. If we were to plot the phase space probability distribution (known as the *Wigner function* in quantum mechanics), you would get a 3D Gaussian "blob" sitting on the 2D $\hat{p} - \hat{x}$ plane.
Now, for linear Gaussian systems, the effect of perturbation ϵ is simply a *linear displacement* of the mean μ of the Gaussian blob. The covariance matrix σ remains the same. This is the simplest case, so we go with the simplest measurement operator: a *homodyne* measurement operator. It has the form

$$\hat{M}_\tau(N) = \frac{1}{\sqrt{2}}\Big(e^{-i\phi}\hat{\mathcal{B}}_\tau(N) + e^{i\phi}\hat{\mathcal{B}}_\tau^\dagger(N)\Big) \tag{57}$$

This operator basically takes the 2D phase space distribution, and projects it onto a 1D line at angle ϕ to the $\hat{x}$-axis.
Another factor that supports our decision to use the homodyne measurement is taking the *large drive limit* ($|\beta| \to \infty$). At finite $|\beta|$, there may still be some dependence of σ on ϵ, due to some fluctuations. Now, the mean μ is proportional to $|\beta|$, as you can understand the center of the Gaussian being dependent on the drive amplitude. However, since the dependence of σ on ϵ is only from fluctuations, it is independent of $|\beta|$. Therefore, if we take $|\beta| \to \infty$ limit, then the μ displacement dominates over changes in σ, so our decision to take homodyne measurement is justified.

1.1.6 Using the measurements

Now, the probability distribution of the measurement outcome m, given parameter ϵ will be Gaussian with

- Mean: $\langle \hat{M}_\tau(N) \rangle_\epsilon$
- Variance: $\mathcal{N}_\tau^2(N, \epsilon) = \langle \hat{M}_\tau^2(N) \rangle_\epsilon - \langle \hat{M}_\tau(N) \rangle_\epsilon^2$

To be clear, $\langle \cdot \rangle_z$ refers to the expectation value of the argument with respect to a state whose dynamics are governed by the Hamiltonian $\hat{H}[z]$.
Now, we've learnt that the Fisher Information is given by

$$I(\theta) = E_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X|\theta) \right)^2 \right] \tag{58}$$

In our case, this becomes

$$I(\epsilon) = \left\langle \left(\frac{\partial}{\partial \epsilon} \log f(m|\epsilon) \right)^2 \right\rangle_\epsilon \tag{59}$$

which we need to evaluate.
Start with the PDF

$$f(m|\epsilon) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(- \frac{(m-\mu)^2}{2\sigma^2} \right) \tag{60}$$

$$\Rightarrow \log f(m|\epsilon) = -\frac{1}{2} \log\left(2\pi\sigma^2\right) - \frac{(m-\mu)^2}{2\sigma^2}$$

$$\Rightarrow \frac{\partial}{\partial \epsilon} \log f(m|\epsilon) = -\frac{\sigma'}{\sigma} - \frac{(m-\mu)\mu'}{\sigma^2} + \frac{(m-\mu)^2 \sigma'}{\sigma^3} \tag{61}$$

where $\mu' = \partial\mu/\partial\epsilon$ and $\sigma' = \partial\sigma/\partial\epsilon$.
(μ and σ represent the mean and variance(noise) of the measurements, which we defined at the start of this section)
Squaring and taking expectation value,

$$I(\epsilon) = \frac{(\sigma')^2}{(\sigma)^2} + \frac{(\mu')^2}{\sigma^4} \langle (m-\mu)^2 \rangle_\epsilon + \frac{(\sigma')^2}{\sigma^6} \langle (m-\mu)^4 \rangle_\epsilon - 2\frac{(\sigma')^2}{\sigma^4} \langle (m-\mu)^2 \rangle_\epsilon + (\text{cross terms with odd moments}) \tag{62}$$

We know that odd moments of Gaussian distribution are zero. We also know that

$$\langle (m-\mu)^2 \rangle_\epsilon = \sigma^2 \qquad \text{and} \qquad \langle (m-\mu)^4 \rangle_\epsilon = 3\sigma^4 \tag{63}$$

Applying these, we get

$$\begin{aligned} I(\epsilon) &= \frac{(\sigma')^2}{\sigma^2} + \frac{(\mu')^2}{\sigma^4}\sigma^2 + \frac{(\sigma')^2}{\sigma^6} 3\sigma^4 - 2\frac{(\sigma')^2}{\sigma^4}\sigma^2 \\ &= \frac{(\sigma')^2}{\sigma^2} + \frac{(\mu')^2}{\sigma^2} + 3\frac{(\sigma')^2}{\sigma^2} - 2\frac{(\sigma')^2}{\sigma^2} \\ I(\epsilon) &= \frac{(\mu')^2}{\sigma^2} + 2\frac{(\sigma')^2}{\sigma^2} \end{aligned} \tag{64}$$

We mentioned in the last paragraph of Section 1.1.5 that the mean depends on $|\beta|$ and the variance doesn't. While the means and variances here are different from the ones in Section 1.1.5 – representing the measurement outcomes *vs* representing the states – we can still see how this dependence holds true for both, since we are measuring states that have this property.
Thanks to this, the first term of (64) dominates over the second term and we get

$$I(\epsilon) \approx \frac{(\mu')^2}{\sigma^2} \tag{65}$$

Thus the QFI for this measurement looks like

$$\mathrm{QFI}_\tau(N) = \left(\frac{\partial_\epsilon \langle \hat{M}_\tau(N)\rangle_\epsilon}{\mathcal{N}_\tau(N,\epsilon)}\right)^2 \tag{66}$$

Now, we define the signal and noise associated with the measurement as follows:

$$\mathcal{S}_\tau(N,\epsilon) = |\langle \hat{M}_\tau(N)\rangle_\epsilon - \langle \hat{M}_\tau(N)\rangle_0| \tag{67}$$

$$\mathcal{N}_\tau(N,\epsilon) = \sqrt{\langle \hat{M}_\tau^2(N)\rangle_\epsilon - \langle \hat{M}_\tau(N)\rangle_\epsilon^2} \tag{68}$$

Now looking at the numerator of the RHS of (66), we can write it as

$$\begin{aligned}\partial_\epsilon \langle \hat{M}_\tau(N)\rangle_\epsilon &= \lim_{\epsilon\to 0} \frac{\langle \hat{M}_\tau(N)\rangle_\epsilon - \langle \hat{M}_\tau(N)\rangle_0}{\epsilon} \\ &= \lim_{\epsilon\to 0} \frac{\mathcal{S}_\tau(N,\epsilon)}{\epsilon}\end{aligned} \tag{69}$$

We can ignore the modulus signs, because we are squaring it in the end anyway. Subsituting this into the QFI expression,

$$\mathrm{QFI}_\tau(N) = \lim_{\epsilon\to 0}\left(\frac{1}{\epsilon}\frac{\mathcal{S}_\tau(N,\epsilon)}{\mathcal{N}_\tau(N,\epsilon)}\right)^2 \tag{70}$$

But we're not done yet. The angle ϕ of our projection can be chosen by us to get maximum QFI. Thus, the final expression is

$$\mathrm{QFI}_\tau(N) = \max_\phi\left[\lim_{\epsilon\to 0}\left(\frac{1}{\epsilon}\frac{\mathcal{S}_\tau(N,\epsilon)}{\mathcal{N}_\tau(N,\epsilon)}\right)^2\right] \tag{71}$$

From this, we can see how the QFI related to the SNR of our optimal homodyne measurement. So instead of aiming to calculate the QFI, we can aim to calculate the SNR and that will be sufficient. A couple points from this:

- From our earlier discussion regarding dependence on drive amplitude $|\beta|$, we can say that the QFI scales as $|\beta|^2$. This of course doesn't imply any true enhancement, you are simply pushing more power into the system.
- Let us taylor expand the signal and noise functions with respect to ϵ in (71). Keeping in mind that, by its definition in (67), $\mathcal{S}_\tau(N,0) = 0$, we expand it to get

$$\begin{aligned}\mathrm{QFI}_\tau(N) &= \max_\phi\left[\lim_{\epsilon\to 0}\left(\frac{1}{\epsilon}\frac{s_1\epsilon + s_2\epsilon^2 + \ldots}{\mathcal{N}_\tau(N,0) + n_1\epsilon + n_2\epsilon^2}\right)^2\right] \\ &= \max_\phi\left[\left(\frac{s_1}{\mathcal{N}_\tau(N,0)}\right)^2\right]\end{aligned} \tag{72}$$

  Thus, you can see that *QFI only depends on Signal upto first order, and Noise upto zeroth order in* ϵ. This will be immensely important going forward.
- Some other factors we are yet to address are the drive phase θ and the perturbation operator $\hat{V}$. We will be interested in optmising those as well.

1.1.7 Performance Metric for Sensors

While it is tempting to treat QFI as the only performance metric for a sensor, one must also remember that different sensors use resources differently. In our case, we wish to distinguish a *true* sensing enhancement from a more trivial effect, such as the QFI increasing simply because more photons were available in the system. For this reason, our performance metric will be the *QFI scaled by the total average photon number* $\bar{n}_{tot}$. We choose to scale it by $\bar{n}_{\mathrm{tot}}$ because photon number is an accurate measure of the resources the system uses. This value is given by

$$\bar{n}_{\mathrm{tot}} \equiv \sum_n \langle \hat{a}_n^\dagger \hat{a}_n\rangle_0 \simeq \sum_n \langle \hat{a}_n^\dagger\rangle_0\langle \hat{a}_n\rangle_0 \propto \frac{|\beta|^2}{\kappa} \tag{73}$$

The approximation is valid in the large drive limit ($|\beta|\to\infty$) as the other terms are not dependent on $|\beta|$ and thus can be ignored.

The last statement of proportionality comes from taking the steady-state case of the equation of motion (41), which means taking $\dot{\tilde{a}}_1 = 0$. Then, taking expectation value on both sides, we get

$$|\langle \tilde{a}_1 \rangle| = |\langle \hat{a}_1 \rangle| \propto \frac{|\beta|}{\sqrt{\kappa}}$$

which gives us the proportionality in (73). From this proportionality, we see that using QFI/$|\beta|^2$ would also be a valid metric, which makes sense as we would be scaling it by drive power.
So our performance metric will be QFI/$\bar{n}_{\text{tot}}$

1.1.8 Computing SNR (setup)

Now our next step is to calculate this optimal QFI. From (71), we see that by computing the SNR, we can easily get the QFI, so that will be our goal.
We start by looking at the signal definition (67). From the definition of the measurement operator in (57), we can get

$$\begin{aligned}\langle \hat{M}_\tau(N) \rangle &= \frac{1}{\sqrt{2}} \Big(\langle e^{-i\phi} \hat{\mathcal{B}}_\tau(N) \rangle + \langle e^{i\phi} \hat{\mathcal{B}}_\tau^\dagger(N) \rangle \Big) \\ &= \sqrt{2}\, \text{Re}\big[e^{-i\phi} \langle \hat{\mathcal{B}}_\tau(N) \rangle \big] \end{aligned} \tag{74}$$

$$\begin{aligned} \Rightarrow \langle \hat{\mathcal{B}}_\tau(N) \rangle &= \frac{1}{\sqrt{\tau}} \int_0^\tau dt\, \langle \hat{B}^{(\text{out})}(t) \rangle \\ &= \frac{1}{\sqrt{\tau}} \int_0^\tau dt \left(\beta + \sqrt{\kappa}\, \langle \hat{a}_1(t) \rangle \right) \\ &= \frac{\beta\, \tau}{\sqrt{\tau}} + \frac{\sqrt{\kappa}}{\sqrt{\tau}} \int_0^\tau dt\, \langle \hat{a}_1(t) \rangle \\ \langle \hat{\mathcal{B}}_\tau(N) \rangle &= \beta \sqrt{\tau} + \frac{\sqrt{\kappa}}{\sqrt{\tau}} \int_0^\tau dt\, \langle \hat{a}_1(t) \rangle \end{aligned} \tag{75}$$

Recall that $\langle \hat{B}^{(\text{in})}(t) \rangle = 0$ since that operator represents noise in (53).
Now, we substitute this in (74) to get

$$\begin{aligned} \langle \hat{M}_\tau(N) \rangle &= \sqrt{2}\, \text{Re}\left[e^{-i\phi} \left(\beta \sqrt{\tau} + \frac{\sqrt{\kappa}}{\sqrt{\tau}} \int_0^\tau dt\, \langle \hat{a}_1(t) \rangle \right) \right] \\ &= \sqrt{2} \cos(\phi) \beta \sqrt{\tau} + \sqrt{\frac{2\kappa}{\tau}} \text{Re}\left[e^{-i\phi} \int_0^\tau dt\, \langle \hat{a}_1(t) \rangle \right] \end{aligned} \tag{76}$$

Now substituting this into the signal definition (67), the first term will vanish due to subtraction, so we are left with

$$\mathcal{S}_\tau(N, \epsilon) = \sqrt{\frac{2\kappa}{\tau}} \left| \text{Re}\left[e^{-i\phi} \int_0^\tau dt \left(\langle \hat{a}_1(t) \rangle_\epsilon - \langle \hat{a}_1(t) \rangle_0 \right) \right] \right| \tag{77}$$

Since ϵ is very small, the signal present due to ϵ is also very small, which is why we need to integrate over a large time interval τ to get enough information. Thanks to this large interval, we can ignore any transient behaviour and safely assume that the system is in steady state (ss). So our equation simplifies to

$$\mathcal{S}_\tau(N, \epsilon) = \sqrt{2\kappa\tau} \left| \text{Re}\left[e^{-i\phi} \left(\langle \hat{a}_1 \rangle_\epsilon^{\text{ss}} - \langle \hat{a}_1 \rangle_0^{\text{ss}} \right) \right] \right| = \sqrt{2\kappa\tau}\, |\text{Re}[e^{-i\phi} \delta \langle \hat{a}_1 \rangle^{\text{ss}}]| \tag{78}$$

where $\delta \langle \hat{a}_1 \rangle^{\text{ss}}$ can be written as

$$\delta \langle \hat{a}_1 \rangle^{\text{ss}} = \langle \hat{a}_1 \rangle_\epsilon^{\text{ss}} - \langle \hat{a}_1 \rangle_0^{\text{ss}} = \epsilon \lim_{\epsilon \to 0} \left(\frac{\langle \hat{a}_1 \rangle_\epsilon^{\text{ss}} - \langle \hat{a}_1 \rangle_0^{\text{ss}}}{\epsilon} \right) + \mathcal{O}(\epsilon^2) \tag{79}$$

Since we only care about signal to the first order in ϵ, we can ignore the higher order terms.
It will be interesting to think of the dynamics in the quadrature picture. For this, we need to bring back the *susceptibility* functions we talked about in the last paragraph of Section 1.1.1. In the quadrature case, it will be a susceptibility *matrix* with four elements $\chi^{\alpha\beta}[n, m; \omega]$. This gives the response of the α quadrature on site n to a driving force of frequency ω

which directly drives the β quadrature at site m. Comparing the equations of motions of the quadratures (10) and (11) to that of the Hatano-Nelson model (2), we see from the similarities that we can use (5) to get the relations

$$\chi^{xx}[n,m;\omega] = e^{A(n-m)}\tilde{\chi}^{xx}[n,m;\omega] \tag{80}$$

$$\chi^{pp}[n,m;\omega] = e^{-A(n-m)}\tilde{\chi}^{pp}[n,m;\omega] \tag{81}$$

where $\tilde{\chi}^{\alpha\beta}[n,m;\omega]$ represents the susceptibility for the equivalent system, just with $A = 0$. Note that these equations are for the case without perturbation, i.e., $\epsilon = 0$. Thus, the two quadratures are uncoupled, and we can also say

$$\chi^{xp}[n,m;\omega] = \chi^{px}[n,m;\omega] = 0 \tag{82}$$

These equations simply confirm what we had thought at the start: $\hat{x}$-quadrature signals are amplified as they travel to the right and deamplified as they travel to the left, while the opposite is true for $\hat{p}$-quadrature signals. This is exactly what we wanted – two opposite Hatano-Nelson chains.

1.1.9 Choices to make

Before we proceed with computing the SNR, we have a few more parameters and operators we need to choose values and forms for. First, we need to decide on the form of our perturbation operator $\hat{V}$.

ASIDE: Non-Hermitian Skin Effect (NHSE) [3] and its corresponding perturbation
This is not particularly important for the rest of the paper, so it won't be as detailed. Refer [3] for more information.
According to [3], the unperturbed Hatano-Nelson chain exhibits NHSE, which is to say that its eigenstates "accumulate" at the boundary of the chain. Since the amplitude of the wavefunction grows to the right, you can understand that, if you expand the eigenstate in terms of the position eigenkets, the ones closer to the right boundary will have a larger coefficient than the further ones.
Because of this accumulation at the edge, changes in boundary conditions (for example, from open to periodic boundary conditions) will drastically affect the system, more than in systems where this accumulation is not present.
Now we want our perturbation to disrupt the system as much as possible, so we get a large response to a small perturbation. The best way to do that here would be to perturb the boundary sites. Thus, we choose the perturbation operator

$$\hat{V}_{\text{NHSE}} = e^{i\varphi}\hat{a}_1^\dagger\hat{a}_N + e^{-i\varphi}\hat{a}_N^\dagger\hat{a}_1 \tag{A1}$$

where φ is an arbitrary phase.
Through calculations that are detailed in the "Methods" section of [1], the signal is found to be

$$\mathcal{S}_\tau(N,\epsilon) = \sqrt{8\tau\kappa\bar{n}_N}\frac{\epsilon}{\kappa}|\sin(\varphi-\phi)| \tag{A2}$$

which shows no real gain. It will increase as number of photons increase, but the decided metric $\text{QFI}/\bar{n}_{\text{tot}} \propto \text{SNR}/\sqrt{\bar{n}_{\text{tot}}}$ will not increase.
Another thing to note is that $\hat{V}_{\text{NHSE}}$ *does not break the* $\mathbb{Z}_2$ *symmetry of the system.* We will show this.
$\mathbb{Z}_2$ symmetry is preserved when a system is unchanged after the combined action of these three operations:

- Time reversal (TR): $\dot{\psi}_n^\sigma \to -\dot{\psi}_n^\sigma$
- Spatial inversion (SI): $\psi_n^\sigma \to \psi_{N+1-n}^\sigma$
- Pseudospin inversion (PI): $\sigma \to \bar{\sigma}$ i.e. swapping the two quadratures $\hat{x} \leftrightarrow \hat{p}$

Applying this to $\hat{V}_{\text{NHSE}}$,

$$\hat{V}_{\text{NHSE}} = e^{i\varphi}\hat{a}_1^\dagger\hat{a}_N + e^{-i\varphi}\hat{a}_N^\dagger\hat{a}_1 \xrightarrow{\text{TR}} e^{-i\varphi}\hat{a}_1^\dagger\hat{a}_N + e^{i\varphi}\hat{a}_N^\dagger\hat{a}_1 \xrightarrow{\text{SI}} e^{-i\varphi}\hat{a}_N^\dagger\hat{a}_1 + e^{i\varphi}\hat{a}_1^\dagger\hat{a}_N \xrightarrow{\text{PI}} \text{unchanged} = \hat{V}_{\text{NHSE}} \tag{A3}$$

The operator is invariant under PI because it couples both quadratures equally.
Thus, the Non-Hermitian Skin Effect does not provide any true advantage in sensing.
END OF ASIDE

If we choose a $\hat{V}$ that breaks the symmetry, we can still achieve our goal of enhanced sensing. Consider the operator

$$\hat{V} = \hat{a}_N^\dagger\hat{a}_N \tag{83}$$

so that $\epsilon\hat{V}$ now corresponds to a perturbation only on the last site of the chain. This perturbation definitely breaks the symmetry of the system, from spatial inversion.

Now we want to see how this affects the equation of motion for the last site (it obviously will not affect the other sites). First, we need to convert it to the quadrature notation using (7):

$$\begin{aligned}\hat{V} &= \hat{a}_N^\dagger \hat{a}_N \\ &= \frac{1}{2}(\hat{x}_N - i\hat{p}_N)(\hat{x}_N + i\hat{p}_N) \\ \hat{V} &= \frac{1}{2}\big(\hat{x}_N^2 + \hat{p}_N^2\big)\end{aligned} \tag{84}$$

We calculate the contribution of the perturbation to the Heisenberg equation of motion:

$$\begin{aligned}\dot{\hat{x}}_N|_\epsilon &= i\Big[\frac{\epsilon}{2}\big(\hat{x}_N^2 + \hat{p}_N^2\big), \hat{x}_N\Big] \\ &= i\frac{\epsilon}{2}\Big(\big[\hat{x}_N^2, \hat{x}_N\big] + \hat{p}_N\big[\hat{p}_N, \hat{x}_N\big] + \big[\hat{p}_N, \hat{x}_N\big]\hat{p}_N\Big) \\ &= i\frac{\epsilon}{2}\big(0 - \hat{p}_N i - i\hat{p}_N\big) \\ \dot{\hat{x}}_N|_\epsilon &= \epsilon\,\hat{p}_N\end{aligned} \tag{85}$$

$$\begin{aligned}\dot{\hat{p}}_N|_\epsilon &= i\Big[\frac{\epsilon}{2}\big(\hat{x}_N^2 + \hat{p}_N^2\big), \hat{p}_N\Big] \\ &= i\frac{\epsilon}{2}\Big(\hat{x}_N\big[\hat{x}_N, \hat{p}_N\big] + \big[\hat{x}_N, \hat{p}_N\big]\hat{x}_N + \big[\hat{p}_N^2, \hat{p}_N\big]\Big) \\ &= i\frac{\epsilon}{2}\big(\hat{x}_N i + i\hat{x}_N + 0\big) \\ \dot{\hat{p}}_N|_\epsilon &= -\epsilon\hat{x}_N\end{aligned} \tag{86}$$

These terms are added to their respective equations of motion (10) and (11) for $n = N$. Obviously $\hat{x}_{N+1} = 0$ and $\hat{p}_{N+1} = 0$. Thus, the equations are

$$\boxed{\dot{\hat{x}}_N = Je^A\hat{x}_{N-1} + \epsilon\,\hat{p}_N} \tag{87}$$

$$\boxed{\dot{\hat{p}}_N = Je^A\hat{p}_{N-1} - \epsilon\,\hat{x}_N} \tag{88}$$

We can see that without the perturbation, the two quadratures evolved independent of each other. But this perturbation has coupled them together. You can understand now why the perturbation was shown as a coupling in Figure 1(a).
Now, we've finalised a form for the perturbation operator $\hat{V}$. We have two more parameters to fix: the phase θ of the drive β, and the projection angle ϕ of the homodyne measurement.
We notice that for the $\epsilon = 0$ case, the susceptibilities $\chi^{xp}[n, m; \omega] = \chi^{px}[n, m; \omega] = 0$, which means signals in one quadrature don't cause responses in the other. Thanks to the coupling caused by ϵ, this will not be true for the $\epsilon \neq 0$ case. Thus, whatever non-zero signal is present in one quadrature due to a signal in the other, is purely due to the perturbation. This is more useful to measure, as there is no background signal to filter out. Thus, we make our input drive purely along the $\hat{x}$-quadrature by setting $\theta = 0$, which makes β purely real. Then we measure along the $\hat{p}$-quadrature by setting $\phi = \pi/2$.
In addition to these, we also make the choice that N, the number of lattice sites in the chain, is *odd*. The reason behind this is the *Chiral Symmetry* of the system.

ASIDE: Chiral Symmetry
The condition for chiral symmetry is that there exists an operator $\hat{\Pi}$ that satisfies [4]:

$$\{\hat{\Pi}, \hat{H}\} = 0 \tag{B1}$$

where $\hat{\Pi}$ is an involutory ($\hat{\Pi}^2 = \mathbb{I}$) and Hermitian operator, which in turn makes it unitary as well.
If an operator that satisfies these conditions also anticommutes with the Hamiltonian, then the system is chiral symmetric. Now, taking an eigenstate $|\psi_\varepsilon\rangle$ of the system, let us define another state $|\phi\rangle$, related by

$$|\phi\rangle = \hat{\Pi}|\psi_\varepsilon\rangle \tag{B3}$$

Let's say the eigenvalue associated with $|\psi_\varepsilon\rangle$ is ε. Then we apply $\hat{\Pi}$ on both sides of the eigenvalue equation to get

$$\begin{aligned}\hat{\Pi}\hat{H}|\psi_\varepsilon\rangle &= \hat{\Pi}\varepsilon|\psi_\varepsilon\rangle \\ -\hat{H}\hat{\Pi}|\psi_\varepsilon\rangle &= \varepsilon\hat{\Pi}|\psi_\varepsilon\rangle \\ \hat{H}|\phi\rangle &= -\varepsilon|\phi\rangle\end{aligned} \tag{B4}$$

From this, we see that for every eigenstate with an eigenvalue ε, there exists another eigenstate with eigenvalue $-\varepsilon$.
END OF ASIDE

There will be as many eigenstates as there are sites in the chain (N). Now we know that the eigenstates exist in $\pm\varepsilon$ pairs, we can see that choosing N to be odd leaves 1 eigenstate out. Since that eigenstate also needs to satisfy the condition, the only possible eigenvalue is $\varepsilon = 0$. Thus, choosing N to be odd makes sure that the lattice will have a resonant mode at zero frequency. This will make our signal calculation easier, as we can put $\omega = 0$ when we calculate susceptibility.
Now we are ready to move forward with our SNR calculation.

1.1.10 Computing SNR and QFI (final)

We start by converting the drive Hamiltonian (46) to quadrature basis

$$\begin{aligned}\hat{H}_{\text{drive}} &= -i\sqrt{\kappa}(\beta\,\hat{a}_1^\dagger - \beta^*\hat{a}_1)\\ &= -i\sqrt{\kappa}\,\beta(\hat{a}_1^\dagger - \hat{a}_1)\\ &= -i\sqrt{\kappa}\,\beta(-i\sqrt{2}\,\hat{p}_1)\\ &= -\sqrt{2\kappa}\beta\,\hat{p}_1\end{aligned} \tag{89}$$

Since the drive Hamiltonian contains $\hat{p}_1$, it will only feature in the EOM of the $\hat{x}$-quadrature, which is in agreement with our earlier statement that the drive is purely in the $\hat{x}$-quadrature.
Now let's look back at the expression for the signal (78), in terms of the real part of $e^{-i\phi}\delta\langle\hat{a}_1\rangle^{\text{ss}}$. Since $\phi = \pi/2$, we only care about the imaginary part of $\delta\langle\hat{a}_1\rangle^{\text{ss}}$, which is the $\hat{p}$-quadrature part $\delta\langle\hat{p}_1\rangle^{\text{ss}}/\sqrt{2}$. We can represent the perturbation caused by ϵ to the equations of motion as δf_N^x and δf_N^p, where (in steady state)

$$\delta f_N^x = +\epsilon\langle\hat{p}_N\rangle_0^{\text{ss}} \qquad \text{and} \qquad \delta f_N^p = -\epsilon\langle\hat{x}_N\rangle_0^{\text{ss}} \tag{90}$$

Recall that $\chi^{\alpha\beta}[n,m;\omega]$ gives the response of the α quadrature on site n to a force driving the β quadrature at site m. Thus, the first-order response in the $\hat{p}$-quadrature is

$$\begin{aligned}\delta\langle\hat{p}_1\rangle^{\text{ss}} &= \chi^{px}[1,N;0]\,\delta f_N^x + \chi^{pp}[1,N;0]\,\delta f_N^p\\ &= \epsilon\chi^{px}[1,N;0]\,\langle\hat{p}_N\rangle_0^{\text{ss}} - \epsilon\chi^{pp}[1,N;0]\,\langle\hat{x}_N\rangle_0^{\text{ss}}\end{aligned} \tag{91}$$

Now that we've factored out the influence of ϵ, χ^{px} is back to being 0, so we get

$$\delta\langle\hat{p}_1\rangle^{\text{ss}} = -\epsilon\chi^{pp}[1,N;0]\,\langle\hat{x}_N\rangle_0^{\text{ss}} \tag{92}$$

Now in steady state, $\langle\hat{x}_N\rangle_0^{\text{ss}}$ is determined by the driving force we obtained earlier. So we get

$$\langle\hat{x}_N\rangle_0^{\text{ss}} = \chi^{xx}[N,1;0]\sqrt{2\kappa}|\beta| \tag{93}$$

$$\Rightarrow \delta\langle\hat{p}_1\rangle^{\text{ss}} = -\epsilon\chi^{pp}[1,N;0]\chi^{xx}[N,1;0]\sqrt{2\kappa}|\beta| \tag{94}$$

There may be some overall phase, but since we eventually taking the absolute value, we can ignore it.
Substituting this into (78), we get

$$\begin{aligned}\mathcal{S}_\tau(N,\epsilon) &= \sqrt{2\kappa\tau}\left|\frac{\delta\langle\hat{p}_1\rangle^{\text{ss}}}{\sqrt{2}}\right|\\ &= \sqrt{\kappa\tau}\sqrt{2\kappa}|\beta|\Big|\chi^{pp}[1,N;0]\,\epsilon\,\chi^{xx}[N,1;0]\Big|\end{aligned} \tag{95}$$

We can now use (80) and (81) to get

$$\begin{aligned}\mathcal{S}_\tau(N,\epsilon) &= \sqrt{2\tau}\kappa\,|\epsilon||\beta|\Big|e^{-A(1-N)}\tilde{\chi}^{pp}[1,N;0]e^{A(N-1)}\tilde{\chi}^{xx}[N,1;0]\Big|\\ &= \sqrt{2\tau}\kappa\,|\epsilon||\beta|e^{2A(N-1)}\big|\tilde{\chi}^{pp}[1,N;0]\tilde{\chi}^{xx}[N,1;0]\big|\\ &= \sqrt{2\tau}\kappa\,|\epsilon||\beta|e^{2A(N-1)}\big|\tilde{\chi}^{xx}[N,1;0]\big|^2\end{aligned} \tag{96}$$

We can understand that in the absence of ϵ, $|\tilde{\chi}^{pp}[1,N;0]| = |\tilde{\chi}^{xx}[N,1;0]|$. Now, we define the average photon number at site N, upto leading order in $|\beta|$.

$$\begin{aligned}
\bar{n}_N &= |\langle \hat{a}_N^\dagger \hat{a}_N \rangle_0| = |\langle \hat{a}_N \rangle_0^{\text{ss}}|^2 \\
&= \frac{1}{2}|\langle \hat{x}_N \rangle_0^{\text{ss}} + i\langle \hat{p}_N \rangle_0^{\text{ss}}|^2 = \frac{1}{2}|\langle \hat{x}_N \rangle_0^{\text{ss}} + i0|^2 \\
&= \frac{1}{2} 2\kappa|\beta|^2 |\chi^{xx}[N,1;0]|^2 = \kappa|\beta|^2|\chi^{xx}[N,1;0]|^2 \\
&= \kappa|\beta|^2 e^{2A(N-1)}|\tilde{\chi}^{xx}[N,1;0]|^2
\end{aligned} \tag{97}$$

Now, we adjust our signal equation to bring in $\bar{n}_N$:

$$\begin{aligned}
\mathcal{S}_\tau(N,\epsilon) &= \sqrt{2\tau}\kappa\,|\epsilon||\beta| e^{2A(N-1)} \left|\tilde{\chi}^{xx}[N,1;0]\right|^2 \\
&= \sqrt{2\kappa\tau}\,|\epsilon|\, e^{A(N-1)} \left|\tilde{\chi}^{xx}[N,1;0]\right| \sqrt{\kappa|\beta|^2 e^{2A(N-1)} \left|\tilde{\chi}^{xx}[N,1;0]\right|^2} \\
&= \sqrt{2\kappa\tau\,\bar{n}_N}\,|\epsilon|\, e^{A(N-1)} \left|\tilde{\chi}^{xx}[N,1;0]\right|
\end{aligned} \tag{98}$$

Now we need some way of obtaining an expression for $\left|\tilde{\chi}^{xx}[N,1;0]\right|$. Through a derivation detailed in the "Methods" section of [1], specifically its Eq.(96), we obtain the expression

$$\tilde{\chi}[n,1;\omega] = i^n \frac{U_{N-n}\left(\frac{\omega}{2J}\right)}{JU_N\left(\frac{\omega}{2J}\right) + i\frac{\kappa}{2}U_{N-1}\left(\frac{\omega}{2J}\right)} \tag{99}$$

where $U_k(x)$ is a Chebyshev polynomial of the second kind $\big(U_k(\cos\theta) = \sin((k+1)\theta)/\sin\theta\big)$+. This equation applies to both $\tilde{\chi}^{xx}$ and $\tilde{\chi}^{pp}$. In our case, we take $\tilde{\chi}^{xx}$, put $n = N$ and $\omega = 0$ to get

$$\tilde{\chi}^{xx}[N,1;0] = i^N \frac{U_0(0)}{JU_N(0) + i\frac{\kappa}{2}U_{N-1}(0)} \tag{100}$$

Using the properties that $U_0(x) = 1$ and $U_k(0)$ is 0 for odd k and $(-1)^{k/2}$ for even k, keeping in mind that N is odd, our expression simplifies to

$$\begin{aligned}
\tilde{\chi}^{xx}[N,1;0] &= i^N \frac{1}{0 + i\frac{\kappa}{2}(-1)^{(N-1)/2}} \\
&= i^{N-1}\frac{2}{\kappa}(-1)^{-(N-1)/2} \\
\Rightarrow |\tilde{\chi}^{xx}[N,1;0]| &= \left|\frac{2}{\kappa}\right|
\end{aligned} \tag{101}$$

Substituting this into our signal expression, we get

$$\mathcal{S}_\tau(N,\epsilon) = \sqrt{8\kappa\tau\,\bar{n}_N}\left|\frac{\epsilon}{\kappa}\right| e^{A(N-1)} \tag{102}$$

Now, for reasons similar to the ones behind NHSE, you can understand that the average number of photons at site N will be exponentially larger than at other sites, especially for large A. We could just approximate it to be equal to $\bar{n}_{\text{tot}}$, but let us say $\bar{n}_N = Z(A)\,\bar{n}_{\text{tot}}$, where $Z(A) = 1 - \mathcal{O}(e^{-4A})$. As a personal note, I don't understand where this correction term comes from, exactly. One can use the fact that $\bar{n}_n \propto |\chi^{xx}[n,1;0]|^2 \propto e^{2A(n-1)}$ to say

$$\bar{n}_{\text{tot}} \propto \sum_{n=1}^{N} \bar{n}_n \propto \sum_{n=1}^{N} e^{2A(n-1)} \propto \bar{n}_N \sum_{n=1}^{N} e^{-2A(N-n)} = \bar{n}_N(1 + e^{-2A} + e^{-4A} + ...)$$

From this, one would think that there would be a $\mathcal{O}(e^{-2A})$ correction factor, but it's $\mathcal{O}(e^{-4A})$ instead, which I can't quite understand. Regardless, the point remains that there is a small correction factor.
Using the relation between the photon numbers, the expression for the signal becomes

$$\boxed{\mathcal{S}_\tau(N,\epsilon) = \sqrt{8Z(A)\kappa\tau\,\bar{n}_{\text{tot}}}\left|\frac{\epsilon}{\kappa}\right| e^{A(N-1)}} \tag{103}$$

(You can see the vague similarities between this and the signal equation for NHSE, given by (A2)*. The method used here is also used there.)*

(103) is the central result of this paper. It shows that even when the photon number is fixed, the signal power will increase *exponentially* with system size. For a visual representation, refer Figure 2 and its caption.

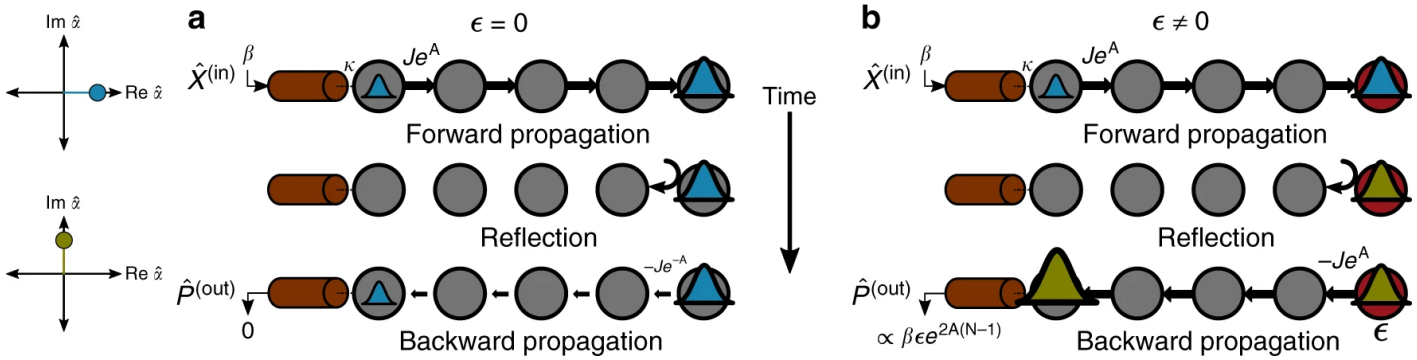


Figure 2: (a) A classical drive is injected into the leftmost lattice site via a waveguide (coupling rate κ). The drive amplitude is real (blue wavepacket), corresponding to an $\hat{x}$-quadrature excitation. As the wavepacket propagates rightwards, its amplitude grows $\propto e^{A(N-1)}$ until it reaches the last site N. If $\epsilon = 0$, the wavepacket scatters off the open boundary without changing quadrature. It is thus deamplified as it propagates back to the first site. As a result, for $\epsilon = 0$, there is no amplification of the drive or of injected noise. (b) For non-zero ϵ, a wavepacket can scatter off the boundary and change quadrature (olive wavepacket). It then is also amplified as it propagates back to the waveguide, and leaves the waveguide with a net amplification factor $e^{2A(N-1)}$. The result is a signal which grows exponentially with system size even when the total intra-cavity photon number is held fixed.

Now, to finish calculating our SNR, we need to examine this system's noise properties. After seeing this picture, one might think that the noise would also be amplified in this way. However, this is not the case. Recall Eq.(72), which shows that the SNR, and thus the QFI, only depends on noise *upto the zeroth order in* ϵ. Without the perturbation, the two Hatano-Nelson chains are decoupled. So the zeroth-order noise that we care about will go through the decoupled chain. It will be amplified and then equally deamplified, as in Fig.2(a).
So the amount of noise in the output field will be equal to the amount of noise in the input field, which we now need to derive. If we split the Gaussian noise operator $\hat{B}^{(\text{in})}$ from (53) into quadratures, we only need to focus on the $\hat{x}$-quadrature, since that is where the input is coming from. Then we find variance $\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 = \langle \hat{x}^2 \rangle$, since mean noise will of course be 0.

$$\begin{aligned}
\hat{x}^2 &= \frac{(\hat{a} + \hat{a}^\dagger)^2}{2} = \frac{\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a}}{2} \\
&= \frac{\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{a}^\dagger \hat{a} + 1}{2} \qquad (\text{using } [\hat{a}, \hat{a}^\dagger] = 1) \\
\Rightarrow \langle \hat{x}^2 \rangle &= \frac{0 + 0 + 2\bar{n}_{\text{th}} + 1}{2} \\
&= \bar{n}_{\text{th}} + \frac{1}{2}
\end{aligned} \tag{104}$$

where $\bar{n}_{\text{th}}$ represents the average number of thermal quanta. These are the photons that are present due to the temperature of the system, not part of the intended drive. Thus, they are the contributors to the noise.
Thus, we declare the noise to be the standard deviation

$$\mathcal{N}_\tau(N, 0) = \sqrt{\bar{n}_{\text{th}} + \frac{1}{2}} \tag{105}$$

Combining these to equations to get the SNR:

$$\text{SNR}_\tau(N, \epsilon) = 4\sqrt{\frac{Z(A)\bar{n}_{\text{tot}}\kappa\tau}{2\bar{n}_{\text{th}} + 1}} \left| \frac{\epsilon}{\kappa} \right| e^{A(N-1)} \tag{106}$$

$$\Rightarrow \boxed{\text{SNR}_\tau(N, \epsilon) = \sqrt{Z(A)} e^{A(N-1)} \, \text{SNR}_\tau(1, \epsilon)} \tag{107}$$

Using (71) to find the QFI, we get

$$\text{QFI}_\tau(N) = 16 \left(\frac{Z(A)\bar{n}_{\text{tot}}\kappa\tau}{2\bar{n}_{\text{th}} + 1} \right) \left| \frac{1}{\kappa^2} \right| e^{2A(N-1)} \tag{108}$$

$$\Rightarrow \boxed{\mathrm{QFI}_\tau(N) = Z(A)\, e^{2A(N-1)}\, \mathrm{QFI}_\tau(1)} \tag{109}$$

And looking at our chosen performance metric,

$$\boxed{\frac{\mathrm{QFI}_\tau(N)}{\bar{n}_{\mathrm{tot}}} = 16\left(\frac{Z(A)\kappa\tau}{2\bar{n}_{\mathrm{th}}+1}\right)\left|\frac{1}{\kappa^2}\right| e^{2A(N-1)}} \tag{110}$$

And therefore, we see the exponential increase in QFI with system size N, which is what we desired.

1.1.11 Comments

Some comments are in order.

- Note how this amplification is not coming from us approaching some unstable point of the system (like the pole of a transfer function). The system is stable throughout the process, for any A and any ϵ (although these equations only apply for small ϵ).
- Since the amplification is exponential, even a few lattice sites can cause a significant increase in QFI.
- One may wonder if this exponential amplification of QFI is in violation of the Heisenberg Limit we established before. That is not the case, as the setting here is different. The usual Heisenberg Limit applies to N-sensor systems, where each one interacts with the parameter of interest. That is clearly not the case here. The enhanced QFI is not a result of entanglement, just a consequence of our system's unusual non-reciprocal amplification.

References

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